

MANVIK TALWAR

AI/ML Engineer | Generative AI, LLM Systems, RAG & Agentic AI
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SUMMARY

AI/ML engineer with three internships shipping LLM systems (local-LLM claims adjudication, enterprise RAG, production APIs). Builds agentic and RAG systems with measured evaluation, deterministic guardrails and human-in-the-loop safety.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Generative AI: LLMs, RAG, Agentic AI, LangGraph, LangChain, Prompt Engineering, Tool Calling, MCP, Embeddings, Vector Search, LLM Evaluation

ML & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face, scikit-learn, CNNs, Computer Vision, OpenCV, MediaPipe, NLP

LLM Inference: llama.cpp, Ollama, GBNF Constrained Decoding, Quantization (GGUF), Azure OpenAI, OpenAI API, faster-whisper

Backend & Data: FastAPI, REST, WebSockets, Pydantic, SQLAlchemy, PostgreSQL, MySQL, MongoDB, Redis, ChromaDB, LanceDB

MLOps & Cloud: Docker, Kubernetes, Prometheus, CI/CD, Git, Linux, pytest, Azure (App Service, Cognitive Search), React, Next.js

PROFESSIONAL EXPERIENCE

AI Engineering Intern | Niva Bupa Health Insurance

May 2026 – Jul 2026

- Built a **claims auto-adjudication engine** for the ReAssure 3.0 health policy: a **7-gate pipeline** (policy, member, coverage, waiting period, exclusion, financial, state update) mixing deterministic rules with a **local Gemma LLM on llama.cpp**.
- Forced schema-valid JSON output with **GBNF grammar-constrained decoding** and built **4-tier confidence routing** that sends uncertain claims to operations, clinical or senior-auditor review queues instead of forcing an automated verdict.
- Built a **DAG rule planner** (Kahn's topological sort) for audit-replayable plans, async **FastAPI** with per-gate **SSE streaming**, **OCR + LLM extraction** from discharge summaries, and a **67-test pytest suite** that runs without a live LLM.

Backend Engineering Intern | TechLearn

Feb 2026 – May 2026

- Shipped the **Dashboard Stats API** serving real-time KPIs to educators, now the **primary analytics endpoint in production**.
- Replaced offset pagination with **cursor-based pagination** on submission retrieval, removing deep-page scan cost; added **composite indexes** and restructured trend queries to cut redundant aggregation passes.

AI Development Intern | Path Infotech

Jun 2025 – Aug 2025

- Deployed an **enterprise RAG assistant** on company servers for internal knowledge retrieval, building the semantic search pipeline with custom **Azure Cognitive Search** indexing skillsets for PDFs.
- Built a **citation layer** backed by MySQL tagging so every generated answer rendered clickable source references, making outputs auditable for non-technical staff.

PROJECTS

IntelliOps: Agentic AIOps Platform | Deloitte Capstone Programme 2026

github.com/CodexManvik/intelliops

- Built a **7-service, event-driven AIOps platform** (Python, FastAPI, Redis Streams, PostgreSQL, Prometheus, React) that collapses alert storms into incidents, ranks root causes and runs **governed, reversible remediation on Kubernetes**.
- On a live cluster: **75% alert-noise reduction**, **41 s median time-to-fix**, 21/30 fixes verified healthy with automatic rollback on failure; autonomy rose **from 0% to 76.7%** of runs as playbooks earned it from verified outcomes. **833 automated tests**.
- Designed **robust anomaly detection** (median/MAD z-scores per metric and hour-of-day, persistence-gated) and confined the **LLM agent to a closed 7-verb action vocabulary** behind a synchronous, fail-closed **human-approval gate** with a full audit trail.

DWT-CNN Invisible Image Watermarking | TensorFlow

github.com/CodexManvik/Deep-Learning-Based-Watermarking

- First-authored an end-to-end differentiable **encode-attack-decode CNN** embedding a 256-bit payload in the Haar DWT LL sub-band; 4.1M parameters, ~18 ms inference per image on an RTX 3050. Manuscript in preparation.
- Made **JPEG differentiable** (straight-through DCT quantization) in a 6-attack simulator; trained on **50K mixed-domain images** to **37.3 dB PSNR**, **0.97 SSIM**, **3.2% BER**, with **2.4–7x lower BER** than a classical DWT baseline.

Agentic RAG for Distributed Content | LangGraph

github.com/CodexManvik/Agentic-Rag-for-Distributed-Content

- Built a **multi-agent LangGraph pipeline** (plan, retrieve, adequacy check, reformulate, synthesize, citation validation) over **hybrid vector + BM25 retrieval**, abstaining instead of hallucinating when evidence is weak or a query is out of policy.
- Wrote an **evaluation harness** (Hit@k, MRR, citation precision, abstain precision/recall); tuned profile reached **0.690 Hit@k** and **0.549 MRR** at 28.8 s median latency on local models and **blocked all prompt-injection test queries**.

Aethel: Local-First AI Agent Runtime | llama.cpp, FastAPI, Tauri, LanceDB

In development

- Local desktop agent with permissioned, rollback-safe tools; **Reflective Skill Memory** keeps or retires skills by success rate.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) | Manipal University Jaipur

Aug 2023 – Aug 2027

ACHIEVEMENTS

- Smart India Hackathon 2025 National Semi-Finalist:** FloatChat-AI, NL-to-SQL and semantic search over ARGO ocean data.
- 4x Student Excellence Award**, Manipal University Jaipur | **Dell x GDG Ideathon Finalist** | **NPTEL Topper**.